Amendment to OSF Registration TDW6C

Late execution of registered validation instruments and replacement human verification of the analytical corpus

Registration: 10.17605/OSF.IO/TDW6C — *Energy performance measurement models in continuous process industries under ISO 50001/50006* (Protocol v2, April 2026)

Document date: 14 October 2026. Date of record: the date of deposit on OSF.

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Amendments of record: (1) Validation Protocol C-2.3, deposited 23 July 2026, OSF file u3wkb; (2) Stage C targeted re-extraction over Corpus B, deposited 13 September 2026, OSF file s8eh5.

Methodological specification: Validation Protocol C-2.3, as departed from by amendment (2) and by this amendment.

Status: Filed before execution and before any sample is drawn; filed after the review's results were first reported (Section 1).

1. Purpose and timing

This amendment records, before execution, the procedures by which part of the declared deviations from the registered design will be remedied before the review is submitted for journal publication. It is filed before any procedure below is executed and before any sample it defines is drawn. The draws are made by a script that refuses to run until the deposit of this amendment has been recorded locally (Section 7).

The amendment is **not prospective with respect to the review's results**. The synthesis has already been reported in the master's dissertation submitted on 30 September 2026 and in a manuscript draft prepared in October 2026. The rules below are therefore fixed before the record-level data they govern are inspected, but not before the aggregate results are known. To limit the latitude this leaves, every frame, sample size, seed, decision category and consequence is fixed here, and every outcome will be reported whatever its direction.

The single-reviewer basis is unchanged; Bernardo Ribeiro is the sole adjudicator of record. **No language model makes, proposes or pre-fills a decision in any procedure below.** Where a stored extraction value is shown to the reviewer, it is the object under verification, and Section 4 states when it is shown. The software that presents items and records decisions is deterministic; it was written with AI assistance and is deposited with the records.

Correction of record. Amendment (2) left its own repository identifier to be recorded after deposit. It is OSF file s8eh5 in project wh62r, deposited on 13 September 2026, SHA-256 `094fd01a8fe3c5c223a8072fa8acaaa216fb9e9a6779c7492288561c59b9e701`. Amendment (1) is OSF file u3wkb, SHA-256 `15c140cdcc18f1f9d9c1dd7520d9ad96741cccafd654ad8b39fac64412ec2650`.

2. Deviations and their treatment

The labels D1–D15 are those of the deviation table in the manuscript. Table 1 states, for each, what this amendment does.

Table 1. Declared deviations and their treatment under this amendment.

No.	Registered design	As executed by 30 Sep 2026	Treatment under this amendment
D1	Screening in three levels (title, abstract, full text)	Title and abstract in one pass	Not reversible; remains declared.
D2	Grey literature searched on reference-organisation websites	Not searched	Remains not performed: it would reopen the corpus (Section 6).
D3	Extraction by a single human reviewer	Model-assisted extraction	Not reversible; partly addressed by Sections 3.3 and 4.
D4	Corpus B extracted by two model families	One model surface	Remains; its purpose is superseded by Section 4.

No.	Registered design	As executed by 30 Sep 2026	Treatment under this amendment
D5	Page locator fixed by the extraction surface	Correctable when the quotation is unique on another page	Unchanged; remains declared.
D6	False-negative control for Corpus B with a reasoned decision per candidate	Partial	Partly addressed by Section 4.1.
D7	Human coverage of Corpus B (registered branch; 11 + 31 under amendment 2)	Not performed; model re-read of 368 cells	Replaced by Section 4. This is a deviation from both designs.
D8	Sub-question SQ4 on applicability to refining	Withdrawn 16 Sep 2026	Remains withdrawn (Section 6).
D9	Backward and forward citation searching	Not performed	Remains not performed: it would reopen the corpus (Section 6).
D10	Random audit of concordant Level-A cells (n = 40)	Not performed	Executed late under the registered rules (Section 3.3).
D11	Reviewer test–retest (n = 30)	Not performed	Executed late; registered escalation adapted (Section 3.2).
D12	Determinism canary (n = 30)	Not performed	Remains not performed: the fixed-configuration condition can no longer be met (Section 3.4).
D13	Terminology search in title/abstract exclusions plus a sample of 50	Not performed	Executed late under the registered rules (Section 3.1).
D14	Yield analysis by search string	Not performed; origin field empty	Reconstructed by re-executing the strings (Section 5).
D15	Rule for deciding that two extractions agree (not specified)	Normalised comparison; rule revised 21 Sep 2026	Unchanged; the revised rule defines the reference values of Section 3.3.

3. Registered instruments executed late

3.1 Title/abstract exclusion audit (D13)

Registered basis. Amendment (1), Section 3 item 2; Validation Protocol C-2.3, Section 3, Step 4b.

Frame. The 326 records labelled *Excluded* at title/abstract screening in the Rayyan export of 24 February 2026 (`data/articles.csv`).

Text assessed. Title and abstract. The retained export carries titles but no abstracts. Abstracts are taken, in this order, from a new Rayyan export of the same review, from OpenAlex (by DOI, otherwise by exact normalised title), or are recorded as unavailable; the source is stored for each record. A record without an abstract is judged on its title and bibliographic data and is counted separately.

(1) **Lexicon census.** The registered continuous-process term list (`output/fn_hunt.py`, `CONTINUOUS_PROCESS_TERMS`, 25 terms, case-insensitive, word-boundary) is run over title and abstract. Every hit is re-checked.

(2) **Random sample.** A simple random sample of 50 records, seed 44, without replacement, from the excluded records with no lexicon hit, the frame sorted by record key before drawing (`random.Random(44).sample`); all are taken if fewer than 50 remain.

Decision. For each record: *exclusion stands* or *should have proceeded to full-text assessment*, judged against the registered screening criteria, with a one-line reason.

Reporting. x/n for the random sample with the exact one-sided 95% upper bound; lexicon hits reported separately, as a census.

Consequences. A record judged *should have proceeded* is sought in full text and assessed against the eligibility criterion. An eligible record is reported by name as a late-identified inclusion, with its sector and whether it meets the Corpus B criterion; it is not charted into the counts of the manuscript, which would require re-running the charting pipeline. If one or more records **of the random sample** are judged *should have proceeded*, the registered escalation option becomes binding: the titles and abstracts of all remaining excluded records are re-screened before submission and the result reported.

3.2 Reviewer test–retest (D11)

Registered basis. Amendment (1), Section 3 item 5 and Section 4.

Frame. The 90 consequential decisions in the 82 blind-first Tier-1 adjudications finalised between 1 and 9 September 2026: 67 on methodological eligibility and 23 on Corpus B membership (`production/reviewer_workspace/reviews/membership/blind`).

Borderline tags. The adjudication form recorded an evidence mode for each decision but no borderline tag, and no decision carries one. Under the registered fill rule the draw is therefore a simple random sample of 30 of the 90 decisions, seed 44, the frame sorted by record key and field.

Procedure. More than five weeks after the original decisions, the reviewer re-decides each sampled decision from the full text alone, with the original decision, its evidence and all model outputs hidden, and records the decision with page and quotation or a *not reported after full-text review* rationale. After commit the original is revealed. A **flip** is any difference between the committed re-decision and the original final decision.

Reporting. The raw flip count x/30 with dates, and the flips listed by record and field.

Escalation. The registered rule (more than one flip triggers re-adjudication of all borderline-tagged decisions) has an empty target set. It is replaced by: every flipped decision is re-adjudicated from the full text and its consequence for Corpus A or B applied and reported; if more than one flip occurs on the same field, all other decisions on that field among the 90 are re-decided under the same procedure.

Limitation. The reviewer has worked with these reports since the original decisions, so recall is possible. Test–retest measures self-consistency, not accuracy.

3.3 Audit of concordant Level-A cells (D10)

Registered basis. Amendment (1), Section 3 item 4 and Section 4; Validation Protocol C-2.3, Section 4a; `output/audit_sampler.py`.

Frame and sample. The 331 Corpus A reports. A simple random sample of 40 reports, seed 44, drawn as in `audit_sampler.select_audit` (frame sorted by key; `random.Random(44).sample(frame, 40)`). Reports are audited in the order returned by the draw, so that any completed prefix is itself a simple random sample.

Fields. The ten Level-A fields — paper type, multi-site, country or region, sector of activity, type of organisation, energy-management standard, M&V protocol, regulatory driver, EnPI/EnB model types and complementary methodologies — with the definitions of the frozen prompts.

Procedure. Blind re-extraction from the PDF with model values hidden until commit. After commit, the reference value of each concordant cell is revealed.

Reference values. The concordant values reported in the manuscript: the two model extractions compared under the rule revised on 21 September 2026 (D15), as recorded in the bibliometric run of that date (`bibliometrics/results/corpus_a_registado/tables/cell_provenance.csv`). Divergent, single-surface and missing cells are excluded from n_v and counted.

Error definition. As registered: enum or boolean, normalised values differ; multi-select, any element added or missing; free text, substantive disagreement, judged and justified; null against a value, an error unless the PDF lacks the datum; *not applicable* against null, an error. After reveal, the reviewer also classifies each difference as an extraction error, a reviewer error found on re-check, or a normalisation-equivalent value. The registered metric is the raw difference count; the adjudicated count is reported beside it and labelled as such.

Reporting. Per field, e_v/n_v with the exact two-sided Clopper–Pearson 95% interval and the one-sided upper bound. Never pooled across fields.

Escalation. As registered, e_v ≥ 3 for a field triggers a corpus-wide recheck of that field and a root-cause note. If a triggered recheck cannot be completed before submission, the field is reported with its audit result, flagged as having

met the threshold, and the recheck reported as pending; the field's results are not presented without that flag. The registered reduction from 40 to 30 reports applies if time runs short; the achieved n and its bound are reported.

3.4 Determinism canary (D12)

Remains not performed. The registered canary measures run-to-run instability under a fixed configuration. The production runs were made in July and August 2026 through consumer applications whose versions and displayed model labels have changed since, so a re-run would be configuration-non-comparable by the registered definition and could not measure what the instrument was designed to measure.

4. Replacement human verification of Corpus B (D7; D6 in part)

This section departs from the registered Corpus B treatment and from the 11 + 31 design of amendment (2), neither of which was executed, and is filed as a deviation.

Frame. The 39 charted Corpus B reports, with the values in force after the model re-read of 17 September 2026 and the author decisions of 18 September 2026 (`stage_c/verification_v1`). The uncharted report rayyan-472536714 is outside every procedure.

Prior exposure. The reviewer has read the extraction forms, the dossiers and the aggregate results for all 39 reports, including the refining and petrochemical cases. No procedure below can be blind-first in the registered sense. Where values are withheld during reading (Section 4.3), this is reported as *values withheld during reading, after prior exposure*, not as blind-first.

4.1 Lexicon-assisted absence census (layer A)

Purpose. To test the absence and rarity statements of the synthesis against the full text of every charted report.

Search. Deterministic, over the page texts from which the extraction was made (`output/stage0/<key>.json`), after lower-casing, joining words hyphenated across line breaks and collapsing white space; case-insensitive regular expressions bounded by word boundaries. The terms are fixed in Table 2 and in the machine-readable file `lexico_camada_A.json` deposited with this amendment (SHA-256 `9f5c4f830a7acc173fe7adc20f4f49f59dc599495f9c9180ebdb68368f1df777`). They are not changed during execution; a term added later is reported as an addition, with its results separately.

Unit and decision. The unit is report × family. For each unit with at least one hit, the reviewer reads every hit at its page in the PDF and records one of: (a) no relevant occurrence; (b) relevant occurrence, already coded; (c) relevant occurrence not coded, with page, quotation, the field affected and the corrected value. Units with no hit are recorded automatically as *no lexicon hit*.

Order. The core families T, F, K, C and Q first; R and V afterwards if time allows, each reported as executed or not.

Consequences. Every (c) corrects the dataset; the affected counts are recomputed, and the manuscript reports the corrected values and the number of corrections per family. **Limitation:** an occurrence expressed without any listed term is not found.

Table 2. Layer A families. Terms are Python regular expressions, each wrapped in word boundaries; `\s` is white space, `?` makes the preceding element optional, `\w*` matches any word ending.

ID	Family (field tested)	Relevant occurrence	Terms
T	Temporal and out-of-sample evaluation (C8 out-of-sample validation)	a model is evaluated on observations not used to fit it; record whether those observations belong to a period after the fitting period (time-blocked test)	`out[\s-]?of[\s-]?sample · hold[\s-]?out · test(?:ing)? (?:set|data|period|sample) · validation (?:set|data|period|sample) · cross[\s-]?validat\w* · k[\s-]?fold · leave[\s-]one[\s-]out · training (?:set|data|period|sample) · calibration period · reporting period · walk[\s-]?forward · rolling[\s-]origin · back[\s-]?test\w* · unseen data · predicted (?:versus|vs\.?) (?:actual|measured|observed) · forecast\w*`

ID	Family (field tested)	Relevant occurrence	Terms
F	Fouling and equipment degradation (C9 mechanisms (cause))	fouling, deposits or equipment degradation is named as a cause of change in the energy relation, of model error or of a model limitation	`foul(?:ing|ed)? · deposit(?:s|ion)? · slagging · soot · scale formation · scaling · cleaning · degradation · deterioration · ag(?:e)?ing · wear`
K	Catalyst activity and run length (C9 mechanisms (cause))	catalyst activity, deactivation, regeneration or run length is named as a cause of change in the energy relation, of model error or of a model limitation	`catalysts? · catalytic · deactivat\w* · regenerat\w* · end[\s-]of[\s-]run · start[\s-]of[\s-]run · run length`
C	Code and data availability (C13 reproducibility)	code or data used for the model is stated to be available (repository, supplementary files, on request)	`github · gitlab · zenodo · figshare · source code · code availability · code (?:is|are) available · data availability · data (?:is|are) (?:openly |publicly |freely)?available · available (?:up)?on (?:reasonable)?request · supplementary (?:materials?|data|information|files?) · supporting information · repository · open[\s-]source`
Q	Feed and product characteristics (C3 variables)	feed or product characteristics (quality, composition, grade, slate, mix) are used or discussed as an explanatory or normalising variable	`feed(?:stock)? (?:quality|composition|type|properties) · feedstocks? · crude (?:oil)?(?:quality|type|slate|blend|mix|assay) · api gravity · raw (?:material|meal) (?:quality|composition|properties) · raw meal · ore grade · moisture content · scrap (?:ratio|share|rate) · product (?:mix|slate|grade|portfolio) · grade changes?`
R[2]	Re-estimation and updating (C6 re-estimation)	a rule or practice for re-estimating or updating the model or baseline is stated	`re[\s-]?estimat\w* · re[\s-]?calibrat\w* · baseline (?:update|updating|revision|adjustment)s? · updat(?:e|ed|ing) (?:the )?baseline · re[\s-]?baselin\w* · re[\s-]?train\w* · re[\s-]?fit\w* · (?:rolling|moving|sliding) window · adaptive · online learning · recursive`
V[2]	Validity domain (C7 validity envelope)	a domain of validity for using the model is stated (operating range, envelope, bounds of applicability)	`validity · valid (?:range|domain|region) · range of (?:validity|application|applicability) · operating (?:range|envelope|window|region) · applicable range · extrapolat\w* · (?:within|outside) the (?:range|domain) · interpolat\w*`

[2] Second priority.

4.2 Sampled verification of result-bearing cells (layer B)

Frame. The 350 cells, of the 368 re-read on 17 September 2026, that remain in the dataset (18 were removed then).

Sample. A simple random sample of 80 cells, seed 44, the frame sorted by item identifier.

Procedure. The reviewer sees the current value, its quotation and page, opens the PDF at that page and reads as much of the report as the value requires, and records: *correct*; *incorrect*, with the correct value; or *cannot be determined*, with the reason.

Reporting. Errors e/n with the exact two-sided Clopper–Pearson 95% interval and the one-sided upper bound, where n excludes *cannot be determined*, which is reported separately; counts by cell type (mechanism, model field, adaptation), descriptively. Errors found are corrected; the rate describes the uncorrected remainder and is reported as such.

4.3 Refining and petrochemical evidence map (layer C)

Frame. The 13 reports whose application was coded as a refinery or petrochemical case and which form the evidence map of the manuscript (Table A1). This set differs from the 11-report head of amendment (2), drawn in sector-priority order from the roster lock, because the procedure verifies what the manuscript reports.

Fields. At report level, the strongest level across the report's models, as in the evidence map:

(1) boundary correspondence of the energy data: exact / allocated (by rule or estimate) / mismatch acknowledged / not reported;

(2) re-estimation regime: declared and executed / declared, not executed / fixed or not stated;

(3) validity domain: with numerical bounds / qualitative / not stated;

(4) out-of-sample result: reported / not reported / not applicable;

(5) test on observations from a later period than the fitting data: yes / no / not applicable.

Procedure. Full-text reading with the coded values withheld until commit; for each field the value, page and quotation, or a *not reported after full-text review* rationale. After commit the coded value is revealed; each difference is re-checked against the source and the reviewer's final value recorded, with reveal changes logged.

Reporting. Per field, the differences x/13 between committed and coded values; the final values and the final map; the reveal-change count, which is not a measure of accuracy.

5. Yield by search string (D14)

The retained export does not record which string retrieved each record (`output/run_metadata.json`, `data_limitations`). The two registered strings will be re-executed in Scopus and in the Web of Science Core Collection with the registered limits (publication year after 1995; article, conference paper and review), and the records exported and matched to the 925 screened records by DOI and, failing that, by normalised title. Reported per string and database: records matched among the 925 screened, among the 599 sought for retrieval, in Corpus A and in Corpus B, and the records of the 925 matched by neither string. The re-execution date differs from the original (February 2026); records published after 24 February 2026 are ignored, and indexing changes since then are a declared source of mismatch.

6. Deviations that remain

Grey-literature searching (D2) and citation searching (D9) remain not performed: each would add records that must be screened, charted and verified, which cannot be completed before submission without lowering the verification applied to the existing corpus. Their potential consequences remain declared, including the cross-match of reference lists already reported. SQ4 (D8) remains withdrawn: Section 4.3 verifies the reported evidence map, not transferability. D4 remains, its purpose superseded by Section 4; D12 remains for the reason in Section 3.4. D1, D3, D5 and D15 cannot be reversed and remain declared.

7. Execution, records and reporting

- **Order.** Sections 3.1 and 3.2 first, because they can change corpus membership; then 4.1, 4.2, 3.3, 4.3 and 5.
- **Window.** From the deposit of this amendment to 29 October 2026. A procedure not completed is reported as not completed, with the completed part and its n.
- **Interlock.** The scripts in `level3_extraction/verificacao_2026-10/` refuse to draw samples or present items until `AMENDMENT_FILED.txt` records the deposit date, the OSF identifier of this file and its SHA-256.
- **Records.** Task files with frames, seeds and draws; append-only decision logs with timestamps and time spent per item; script versions (git commit); and the reports. They are deposited in OSF project wh62r before peer review.
- **Reporting.** Each instrument is reported separately; no pooled error rate and no agreement coefficient. Any count that changes replaces the earlier count in the manuscript, with what changed and why, and the manuscript describes these procedures as executed after the results were first reported.

Appendix A. Reports in layer C

Table A1. The 13 reports of the refining and petrochemical evidence map.

Record key	Report
rayyan-472536695	Velázquez et al. (2013)
rayyan-472535680	Yao et al. (2016)
rayyan-472535365	Moghadasi et al. (2025)
rayyan-472536532	Beisheim et al. (2019)
rayyan-472535550	Hamedi and Mokhtar (2019)
rayyan-472535524	Beisheim et al. (2020)
rayyan-472535649	Valencia-Ochoa et al. (2017)
rayyan-472535582	Morales et al. (2018)
rayyan-472535351	Almanza and Leon-De-Los-Santos (2025)
rayyan-472535625	Ikeyama et al. (2017)
rayyan-472536551	Hashim et al. (2019)
rayyan-472536518	Maggiore et al. (2020)
rayyan-472536425	Hashim et al. (2022)